Quantifying Ocular Confounds in EEG Decoding Without an Eye Tracker

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Abstract—Eye movements, blinks, and the corneo-retinal standing potential leak into scalp electroencephalography (EEG) and co-vary with task structure, so a classifier that claims to decode a cognitive state may instead be partly reading the eyes. Existing defenses either remove ocular signal or model it within a single eye-tracked recording; none provides a portable measurement of how much of a published decoding result could be ocular, for the many datasets recorded without an eye-tracker. We define and validate an auditing instrument. A frozen linear EEG-to-gaze readout, trained where synchronized gaze exists, induces a locked ocular subspace (the orthogonal projection onto the span of its weights). The auditor operates on all available channels: we report a full 64-channel montage and, as an explicitly specified subset, the 30-channel ERP-CORE set. The Ocular Predictability Index (OPI) is the cross-validated area under the curve (AUC) of a paradigm-matched decoder applied to that projection, minus a within-subject label-permutation null. The instrument ships with a three-tier reliability gate that returns "inconclusive" when gaze cannot be reconstructed in the target montage, a measured two-sided bias envelope, and a computable neural-leakage bound. Validating against ground truth in a physically grounded simulator of EEG with synchronized gaze and on real 64-channel EEG (PhysioNet eegmmidb), the OPI is 0.43 on saccade-coupled ocular data and 0.20 on a condition-correlated eye-blink artifact, versus at most 0.03 on every negative control, rising monotonically with the artifact-label coupling. A lateralized neural source decodable from full EEG (AUC 0.79) is not decodable from the ocular projection (OPI 0.007, not significant). Neural leakage is governed by the principal-angle overlap between the cognitive topography and the ocular subspace (Spearman correlation 0.95). A frozen, montage-matched ocular-subspace auditor thus turns "how much of this is the eyes?" into a calibrated, two-sided number with a self-declared competence boundary, enabling portable ocular-leakage audits of EEG decoding results recorded without an eye-tracker.

Index Terms—EEG decoding, ocular artifact, eye movements, construct validity, confound auditing, brain–computer interface, subspace projection

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I. Introduction

A decoding result is only as trustworthy as the signal it rests on. In EEG, a recurring threat to construct validity is that a classifier credited with reading a cognitive state (attention, working-memory load, target detection, the meaning of a read word) is partly reading the eyes. The eye is an electrical dipole, cornea-positive and retina-negative, with a standing potential of roughly 0.4–1.0 mV [1]; its rotations, together with blinks and the broadband saccadic spike potential at saccade onset [2], project to frontal and periocular electrodes with amplitudes that dwarf cortical signals (Fig. 1). Crucially, the field of an eye movement depends on each electrode's position relative to the eyes: its sign reverses across the midline (a dipole) and its magnitude falls with distance, and a blink imposes a different, symmetric pattern, so the contamination an electrode sees is montage- and artifact-specific. Because where and when a person looks is rarely independent of the experimental condition, this ocular activity is condition-correlated: it can be decoded in lockstep with the cognitive label and inflate apparent accuracy. The danger is the machine-learning failure mode of shortcut learning: exploiting an incidental correlate rather than the intended construct [3].

The standard defenses are incomplete. Independent component analysis and regression remove ocular components [4], [5], but removal is imperfect and leaves a residual whose decodable content is unknown; overweighting the saccadic spike before ICA improves this but still targets removal, not measurement [6], [7]. Deconvolution frameworks rigorously model ocular and overlap effects, but in practice within a single, carefully co-registered, eye-tracked recording [8], [9]. None of these answers the question an author or a reviewer actually faces about a finished result: what fraction of this decodable signal could be ocular? This is especially true for the thousands of

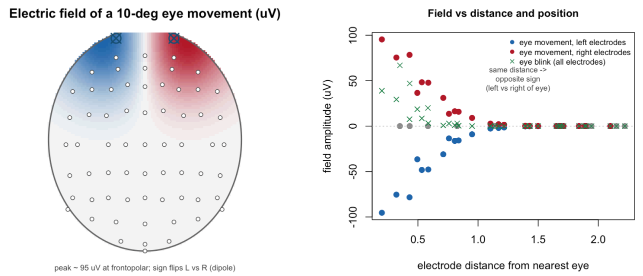

Fig. 1. The electric field of an eye movement depends on electrode position. Left: scalp field of a 10° horizontal eye movement over the 64-channel montage (peak $\approx 95\,\mu$V at frontopolar sites; sign reverses left/right of the eyes, a dipole). Right: field amplitude versus each electrode's distance from the nearest eye. At a given distance the sign depends on which side of the eyes the electrode sits (blue vs. red); an eye blink (green) imposes a distinct, symmetric, one-signed field that also decays with distance. This position- and artifact-dependence is why the auditor is montage-matched.

datasets recorded with no eye-tracker at all.

The concern is not hypothetical. Mostert et al. [10] showed that a visual working-memory "representation" could be partly explained by systematic eye movements, and constructed a bespoke, within-experiment control: if the target variable cannot be decoded from simultaneously recorded gaze, the neural result is safer. Rotaru et al. [11] found that EEG decoders of the spatial focus of auditory attention overfit to eye-gaze and per-trial fingerprints, dropping to chance once gaze was controlled. More broadly, the neuroimaging and machine-learning literatures increasingly reward auditing the field's methods and benchmarks for inflated inference: cluster-extent false positives in fMRI [12], pervasive data leakage in ML-based science [13] and specifically in deep-learning EEG [14], and permutation-based tests for confounding [15]. What is missing is a transferable instrument: a frozen, montage-matched ocular-subspace probe that can be pointed at an arbitrary EEG decoding dataset (including one with no gaze recording) and that returns a calibrated number together with an honest statement of when it cannot be trusted.

This paper defines and validates that instrument. Our contributions are: (i) a locked, linear ocular-subspace operator and an Ocular Predictability Index (OPI) with a within-subject permutation null and a full-EEG ceiling; (ii) a three-tier reliability gate that returns "inconclusive" rather than a false clean bill of health; (iii) a measured two-sided bias envelope, so the index can neither silently hide contamination nor over-blame the eyes; and (iv) a computable neural-leakage bound: the one residual failure mode, that the projection absorbs genuine neural signal, is shown to be governed by the principal angle between the cognitive topography and the ocular subspace, converting a reviewer's objection into a number. We validate all four against ground truth in a physically grounded simulator and demonstrate the instrument on real 64-channel EEG.

II. Methods

A. Design and rationale

The instrument is deliberately linear. Although a deep network is one way to learn an EEG-to-gaze readout where synchronized eye-tracking is available [16], [17], the object that is frozen and shipped is a linear readout and the orthogonal projection onto the span of its weights. We therefore implement and validate exactly that object. Because ground truth (knowing how much of a decodable signal is truly ocular) is available only when gaze is recorded or simulated, the primary validation uses a forward-model simulator with known sources, complemented by a real-EEG negative control.

The auditor uses every channel a dataset provides; dimensionality is never silently reduced. We report a full 64-channel 10–10 montage as primary and treat the 30-channel ERP-CORE set as an explicitly specified subset of it, used only to characterise how channel count and near-eye frontal coverage affect the auditor (the reliability gate, below). Validation draws on more than one dataset: EEGEyeNet (hundreds of subjects with 128-channel EEG and synchronized eye-tracking) is the source paradigm and montage family from which the gaze readout would be trained [16]; ERP CORE supplies the 30-channel target montage [18]; PhysioNet eegmmidb supplies real 64-channel EEG [19]; and ZuCo provides a same-cap-family reading corpus for the near-fixation regime [17]. Source amplitudes are taken from the electrooculography and saccade literature.

B. The locked ocular-subspace operator

Let $\mathbf{X}_{\text{src}} \in \mathbb{R}^{n\times p}$ be source EEG (trials $\times$ channels) for which ocular ground truth $\mathbf{Y}_{\text{src}} \in \mathbb{R}^{n\times q}$ is available (q outputs: horizontal and vertical gaze, a saccade channel, and a blink indicator). A ridge readout $\mathbf{W} = \arg\min_{\mathbf{W}} \|\mathbf{Y}_{\text{src}} - \mathbf{X}_{\text{src}}\mathbf{W}\|_F^2 + \lambda\|\mathbf{W}\|_F^2$ maps EEG to ocular state. The ocular subspace is the column space of $\mathbf{W}$, and the locked operator is the orthogonal projection onto it,

$$\mathbf{P} = \mathbf{U}\mathbf{U}^\top, \qquad \mathbf{U} = \text{left singular vectors of } \mathbf{W}, \tag{1}$$

a symmetric idempotent $p\times p$ matrix ($\mathbf{P}^2 = \mathbf{P}$). Projecting target EEG, $\mathbf{X} \mapsto \mathbf{X}\mathbf{P}$, retains only the component lying in the ocular readout's span. The operator is fixed by (1) before any audit; the auditor is retrained on each target's exact channel set (montage-matched, no interpolation of periocular channels).

C. The Ocular Predictability Index

A paradigm-matched cognitive decoder (a ridge linear classifier) is trained and tested under strictly subject-wise cross-validation on the projected data $\mathbf{X}\mathbf{P}$, yielding a macro-averaged AUC. The OPI subtracts a within-subject label-permutation null (labels shuffled within each subject, $B = 200$):

$$\text{OPI} = \text{AUC}_{\text{proj}} - \overline{\text{AUC}}_{\text{null}}, \tag{2}$$

and is reported alongside the full-EEG AUC as a ceiling and a permutation p-value. The auditor never sees the target cognitive labels, so audits are out-of-distribution by construction.

D. Reliability gate and bias envelope

Reliability is the cross-validated gaze-reconstruction R^2 of the montage-restricted readout, a property of the montage that requires no target eye-tracker. Thresholds are derived empirically from the relationship between reliability and index bias and partition results into confirmatory, exploratory, and inconclusive tiers. The bias envelope is measured, not assumed: where ground-truth gaze exists, we compare the auditor's OPI to the OPI computed directly from true gaze across the saccade-amplitude range and report the signed deviation as a two-sided margin.

E. The neural-leakage bound

The instrument's central risk is that the projection absorbs genuine neural activity that happens to be gaze-correlated, inflating the index. We bound this. For a target whose cognitive decoder has contrast topography $\mathbf{t}$, define the overlap

$$s = \frac{\|\mathbf{Pt}\|}{\|\mathbf{t}\|} = \cos\theta, \tag{3}$$

the cosine of the principal angle θ between $\mathbf{t}$ and the ocular subspace. s upper-bounds the fraction of the neural topography that can enter the projection; under the linear model, leakage $\to 0$ as $s \to 0$. We report s per audit and test the predicted monotonicity empirically.

F. Grounded EEG+gaze simulator

Ground-truth validation uses a forward-model simulator on a full 64-channel 10–10 montage; the 30 ERP-CORE electrodes [18] are an explicitly labelled subset of these 64, so the auditor is exercised on all 64 channels and the 30-channel case is a deliberately studied restriction, never a silent reduction. Per-trial channel patterns are sums of physically grounded source topographies (Fig. 1): a horizontal corneo-retinal field ($\approx 16\,\mu$V per degree), a vertical/blink frontopolar field, a focal periocular saccadic spike scaling with saccade size [2], an $\approx 80\,\mu$V frontopolar blink, and a lateralized posterior N2pc-like neural component ($\approx 1.8\,\mu$V) [20], on spatially correlated background noise. An artifact is coupled to the cognitive label with tunable strength, implementing the condition-correlated premise. Scenarios include saccade-coupled ocular and blink-coupled ocular (two positive controls testing distinct artifact types), covert attention (lateralized posterior neural source, central gaze), focal motor imagery, label shuffle, and a neutral baseline.

G. Real data

We use PhysioNet eegmmidb (all 64 channels, BCI2000, 160 Hz, EDF+; 8 subjects) [19], [21]; no channels are dropped. Because eegmmidb has no eye-tracker, the ocular subspace is estimated from the eyes-open baseline run of four source subjects (a real-EEG analogue of a frozen gaze-trained auditor) and applied to four held-out target subjects' motor runs. We audit two contrasts: left-versus-right imagined movement (no differential gaze expected; the clean negative control) and rest-versus-movement.

H. Implementation

The auditor, simulator, and analyses are implemented in R (CPU-only). AUC is computed by the rank statistic and cross-checked against the pROC package [22] to $< 10^{-6}$. The trained auditor is small and CPU-deployable; code and results are released. The deployment vision, applying the frozen instrument across public decoding benchmarks (e.g., [18], [23]–[25], accessed via MOABB [26], [27]), is discussed in Section IV.

III. Results

A. The operator behaves as specified

Unit tests confirm the rank-based AUC matches pROC to $< 10^{-6}$ including ties; the projector of (1) is symmetric and idempotent with the correct rank, fixes its own column space, and annihilates the orthogonal complement; and, end-to-end, the OPI is large and significant when the cognitive signal lies in the ocular subspace and non-significant when it lies outside it. The trained auditors had ocular-subspace rank 4 (matching the four ocular outputs) on the simulator montages and rank 2 on real eegmmidb.

B. Graded sensitivity and specificity (H3)

Across 12 independent simulated studies per cell on the full 64-channel montage, the OPI cleanly separates ocular leakage from neural signal (Table I). The two positive controls test distinct artifact types: saccade-coupled ocular gives OPI = 0.429 with projected AUC equal to the full-EEG ceiling (0.929), and a condition-correlated eye-blink artifact gives OPI = 0.198 (projected AUC 0.697), both significant in 100% of studies. The auditor therefore detects blink contamination as well as saccade contamination. Every negative control is near-null in magnitude: motor imagery 0.031, label-shuffle 0.007, covert attention 0.007, neutral −0.003; their envelope sets an empirical calibrated neural-leakage margin of ≈ 0.05 that the index must clear beyond the bare permutation null. Decisively, the covert-attention contrast is decodable from full EEG (ceiling AUC 0.79) yet not from the ocular projection (OPI 0.007, $p = 0.48$): the auditor does not absorb lateralized posterior neural signal.

The index is a graded meter, not a binary flag: as the gaze–label coupling ρ increases from 0 to 1, the OPI rises monotonically and near-linearly from −0.014 to 0.500,

TABLE I
H3 sensitivity battery: the OPI separates ocular leakage from neural signal (mean of 12 simulated studies; full 64-channel montage).

Scenario	OPI	proj	ceiling	% sig
Saccade-coupled ocular[a]	0.429	0.929	0.930	100
Blink-coupled ocular[a]	0.198	0.697	0.781	100
Motor imagery	0.031	0.531	0.842	42
Covert attention[b]	0.007	0.507	0.785	17
Label shuffle	0.007	0.507	0.505	0
Neutral	−0.003	0.496	0.635	0

proj = AUC on the ocular projection; ceiling = full-EEG AUC; % sig = studies with permutation $p < 0.05$.
[a] Positive control (two distinct artifact types).
[b] Neural-leakage test: decodable from full EEG, not from the ocular subspace.

TABLE II
Reliability (R^2), bias, and trust tier across saccade amplitude and montage.

Amp. (deg)	R^2	bias	tier
Full montage (64 ch)			
0.25	0.664	−0.034	confirmatory
1.0	0.899	0.000	confirmatory
12.0	0.755	0.000	confirmatory
ERP-CORE subset (30 ch)			
0.25	0.497	−0.056	exploratory
1.0	0.775	0.000	confirmatory
12.0	0.679	0.000	confirmatory

Thresholds (empirical): confirmatory $R^2 \geq 0.664$, exploratory $R^2 \geq 0.497$. Representative amplitudes shown; full grid in released tables.

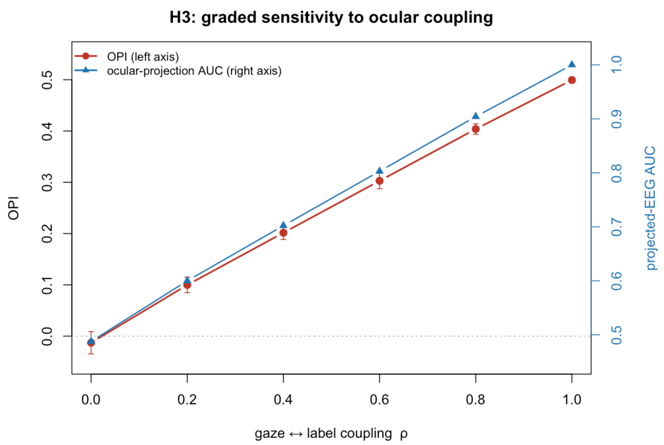

Fig. 2. Graded sensitivity: OPI (red, left axis) and ocular-projection AUC (blue, right axis) rise monotonically with the gaze–label coupling.

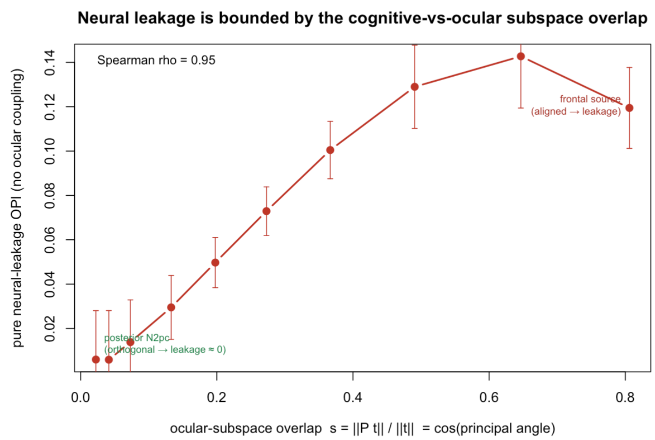

Fig. 3. Neural leakage versus the cognitive-vs-ocular subspace overlap $s = \cos\theta$. Pure neural-leakage OPI rises monotonically with s ($\rho = 0.95$) and is ≈ 0 for a posterior (orthogonal) source.

with the projected AUC tracking from chance to 1.0 (Fig. 2).

C. The bias envelope is conservative

Comparing the auditor's OPI to the OPI from ground-truth gaze across the saccade-amplitude range (Table II), the bias is within ± 0.01 from 0.5° upward on both montages. Only at the fixational extreme (0.25°) does it grow, to −0.034 on the full 64-channel montage and −0.056 on the 30-channel subset, and in both cases the auditor understates the ocular contribution, the conservative direction. No over-statement was observed. The breach magnitude is measured and folded into the trust tiers as a two-sided margin.

D. The reliability gate has teeth

The empirically calibrated thresholds are $R^2 = 0.664$ (confirmatory) and $R^2 = 0.497$ (exploratory). Tiering the amplitude × montage grid (Table II), the full 64-channel montage is confirmatory at every amplitude, while the 30-channel subset falls to exploratory at the fixational extreme, precisely the regime (sparse montage, sub-degree movement) where a silent failure would be most dangerous.

E. Neural leakage is bounded by a measurable angle

Sweeping a lateralized neural source from posterior to frontal with no ocular coupling (so any positive OPI is pure leakage), the leakage is governed by the overlap s of (3) (Fig. 3). It rises monotonically with s (Spearman $\rho = 0.95$ across the swept range), from +0.006 at near-orthogonality ($s \approx 0.02$, the posterior-N2pc regime) to +0.12–0.14 when the source is frontal ($s \approx 0.6$–0.8). The headline posterior case is thus low-leakage under the linear model, and any target with a frontal cognitive topography is flagged by a large s for the leakage-corrected envelope.

F. A headline-analog audit is gated honestly

To emulate the deployment case we audited a simulated ERP-CORE-like N2pc target on the sparse montage carrying a sub-0.2° condition-correlated ocular residual of the kind that survives strict eye-movement rejection. The

TABLE III
Real-EEG audit (eegmmidb; ocular subspace frozen from source subjects' eyes-open baseline; subject-wise evaluation).

Contrast	ceiling	proj	OPI	p
Left vs. right imagery[a]	0.601	0.501	−0.023	0.998
Rest vs. move[b]	0.672	0.615	+0.114	0.002

[a] Clean negative control (no differential gaze): not decodable from the ocular subspace.
[b] Cautionary: a no-eye-tracker channel proxy retains frontal neural signal.

OPI was positive and significant (0.150, ceiling AUC 0.777, $p < 0.005$), but the reliability gate returned "inconclusive" ($R^2 = 0.429$, below the exploratory threshold). The instrument withholds certification of a positive, significant index because the montage and regime cannot reconstruct gaze well enough to trust it: the intended "inconclusive by design" safeguard, not a false alarm.

G. Real-EEG negative control

On real 64-channel eegmmidb with the ocular subspace frozen from independent subjects' eyes-open baseline (Table III), the left-versus-right imagery contrast, where the eyes have no reason to differ, is weakly decodable from full EEG (AUC 0.601) but not from the ocular subspace (OPI −0.023, $p = 0.998$): a clean negative control on real data. The rest-versus-move contrast is included as a cautionary case: a frontal-channel-proxy subspace (the best obtainable without an eye-tracker) retains frontal neural signal it cannot separate from ocular activity (OPI +0.114, $p = 0.002$). This is the empirical argument for the instrument's design: the auditor must be trained on synchronized gaze, which isolates the ocular subspace from frontal neural activity, rather than on channel proxies.

IV. Discussion

We have defined a frozen, montage-matched ocular-subspace auditor and shown, against ground truth, that it does what an audit instrument must: it is sensitive to more than one artifact type (OPI 0.43 for saccade-coupled and 0.20 for blink-coupled contamination), specific (near-null on neural and shuffled controls), graded (monotone in the coupling), conservative (a measured, signed bias envelope that errs toward under-stating), self-aware (a reliability gate that returns "inconclusive"), and, critically, honest about its one residual failure mode, neural leakage, which we render as a computable number, the principal-angle overlap s. Throughout, the auditor uses all 64 channels; the 30-channel ERP-CORE case is reported as an explicit subset to quantify how channel count affects reliability, not as a hidden reduction.

Relation to prior work. The instrument generalizes the within-experiment control of Mostert et al. [10] and the single-domain diagnosis of Rotaru et al. [11] into a transferable meter that requires no eye-tracker on the target, complementing rather than replacing ocular removal [4], [6] and within-recording deconvolution [8]. Where confound-quantification in machine learning probes a measured nuisance variable [15], here the nuisance (gaze) is physically recovered even when it was never recorded in the target, and the back-door through which it could mislead (neural leakage) is bounded by (3). This positions the work in the now-active genre of auditing the field's results for inflated inference [12]–[14].

Toward a field-wide audit. Because the instrument is portable and CPU-cheap, the natural deployment is an ocular-leakage audit across widely used public benchmarks: community-trusted ERP data [18], P300 [28] spellers including overt/covert designs [24], [25], rapid visual streams without any eye-tracker [23], and motor-imagery sets, accessed uniformly via MOABB [26], with each no-eye-tracker benchmark paired with a gaze- or EOG-bearing anchor so that verdicts are ground-truthable. The competence boundary makes such a leaderboard honest: a benchmark on which gaze cannot be reconstructed is reported "inconclusive," not falsely cleared.

A. Limitations

The ground-truth validation is necessarily simulation-based, because measuring bias requires knowing the true ocular contribution; the simulator is grounded in measured physiology but is a simplification (notably, spatial features rather than full spatiotemporal windows, and a linear generative structure aligned with the linear operator). The leakage bound s upper-bounds the overlap and its monotonic relationship to leakage is shown empirically under the linear-subspace model; both should be tested across real cognitive topographies. Real synchronized-gaze datasets such as EEGEyeNet [16] and ZuCo [17], [29] are the appropriate anchors for training the readout and validating verdicts, but their training paradigms are cued large saccades, so the sub-degree fixational regime must be anchored on reading microsaccades rather than assumed. The real-EEG demonstration used a channel-proxy subspace because eegmmidb has no eye-tracker; it therefore validates the negative-control logic and the design rationale, not a gaze-trained transfer. Finally, the index measures only the ocular subspace; it will not detect non-ocular confounds such as repeated-stimulus fingerprints.

V. Conclusion

How much of an EEG decoding result could be the eyes rather than the mind is, for most published datasets, an unanswerable question because no eye-tracker was present. A frozen, montage-matched ocular-subspace auditor makes it answerable: it converts the worry into a calibrated, two-sided Ocular Predictability Index with a self-declared competence boundary and a computable neural-leakage bound, validated here against ground truth and demonstrated on real EEG. The instrument is small, CPU-runnable, and released, and it is built to be pointed at the field's benchmarks, reporting "inconclusive" wherever it honestly cannot see.

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The authors thank the providers of the open datasets used in this work. All datasets analyzed are publicly available from their original repositories (EEGEyeNet, ERP CORE, ZuCo, PhysioNet eegmmidb, and Brain Invaders); the auditor implementation, simulator, and all analysis code and result tables are released with this article. This study used only publicly available, de-identified datasets and synthetic data and involved no new human-subjects data collection. The authors declare no competing interests and received no specific external funding.

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